



Max. Marks: 100

20 Marks

- 1-

11. What does the Lagrange-Euler Formulation in mobile robot dynamics primarily describe?
 A) Sensor performance B) The cost of robot production
 C) The relation between force and its motion D) The color and design of the robot
12. In a given differential wheel drive mobile robot, if the first (left) wheel generating a traction force of 1 N and the second (right) wheel also generating a traction force of 2 N. Then, what will be the vehicle motion?
 A) move forward B) move backward
 C) move forward and rotate counterclockwise D) move forward and rotate clockwise direction
13. The main purpose of deriving the dynamic relations for mobile robots is to:
 A) Design the physical appearance of robots B) Predict the robot's future positions only
 C) Calculate the energy consumption for marketing purposes D) Understand how forces and torques affect the robot's motion
14. What does LiDAR stand for?
 A) Linear Detection and Ranging B) Light Direction and Ranging
 C) Light Detection and Ranging D) Linear Distance and Ranging
15. Choose the false statement:
 A) Prediction update is based on the updated belief state B) Perception update decreases the uncertainty as the robot moves
 C) Action update is based on the prior belief state. D) Prediction update increases uncertainty as the robot moves
16. In Markov localization, The Markov assumption is made which assumes that
 A) Only current sensor measurements are required for accurate localization B) The probability that a robot's state estimate equals the actual robot state is maximized.
 C) The current state is only dependent on the previous state and its most recent actions and perception D) Tracks the robot and is inherently very precise and efficient
17. Which path planning method utilizes a grid of cells to determine the shortest path?
 A) Cell decomposition methods B) Visibility graph
 C) Voronoi diagram D) Potential Field methods
18. The A* Algorithm is best described as:
 A) An obstacle avoidance technique B) A sensor calibration method
 C) A graph search algorithm for finding the shortest path D) A method for robot assembly
19. Which algorithm is specifically designed for obstacle avoidance?
 A) Breadth-first search B) Dijkstra's algorithm
 C) Bug Algorithm D) Grass fire
20. What is the primary purpose of a visibility graph in robot path planning?
 A) To find the shortest path that avoids obstacles. B) To construct a graph that ignores obstacles.
 C) To create a dense network of nodes and edges. D) To navigate without using any sensors.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) What is the degree of mobility, steerability and maneuverability of a mobile robot? Find the degree of Mobility, degree of Steerability and maneuverability for the following wheeled mobile robot configurations. The solid shaded wheels are powered and hatched lines are not powered wheels.

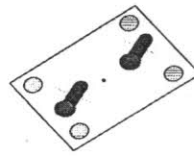
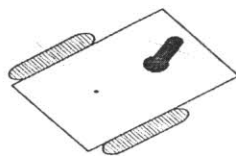
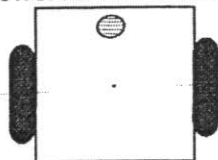


Figure. Q1(a)

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- b) What are the characteristics of an omni-wheel mobile robot?
For a given mobile robot with three omni-directional wheels, if angular velocities are $\begin{bmatrix} 0.5 \\ -0.25 \\ -0.25 \end{bmatrix}$ rad/s, $a=0.2m, l = 0.3m$ find the longitudinal velocity (u) lateral velocity (v) and angular velocity (r) of the robot with respect to the body frame?
2. a) What is forward kinematics of a mobile robot? Derive an expression of the forward kinematics ($\dot{\eta} = J(\psi)\zeta$) for a mobile robot?
- b) The vector of the input velocity command of the land-based mobile robot is given as $\zeta = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$ and the mobile robot (body) frame is oriented 45° with respect to the fixed frame. In this case, what will be the time derivatives of the generalized coordinates?
- c) In a two fixed wheel (differential) drive mobile robot, the left side wheel angular velocity is 1 rad/s and the right-side wheel angular velocity is 1.5 rad/s, both wheels are of same size, and the wheel radius is 10cm. The wheels are placed at an equal distance from the body frame. The distance between the left wheel and right wheel is 40 cm. What will be the longitudinal velocity (u) and angular velocity (r) of the robot with respect to the body frame?
3. a) Describe and compare various types of land-based mobile robots, highlighting their locomotion mechanisms and applications.
- b) Consider a mobile robot equipped with four Mecanum wheels and initially oriented parallel to the x-axis. The robot is required to move clockwise direction about its centre, determine the appropriate combination of wheel angular velocities? Provide a proof to support your answer.
- Unit – II**
4. a) A mobile robot with a tricycle configuration has a powered and steerable wheel at the front. The radius of wheel is 50N, at a steering angle of 30° , then what are the forces/Moments (F_x, F_y, M_z) at the body fixed frame? The distance from the Robot frame to the steering axis, along the positive X axis is to be 0.2m.
- b) Explain the concept of Kalman Filter localization and its application in enhancing the localization accuracy of mobile robots
5. a) Derive dynamic equation of motion for an Omni wheeled mobile robot.
- b) What is Markov localization? Explain using a case study.
6. a) In a given differential wheel drive mobile robot, if the first (left) wheel generates a traction force of 1 N and the second (right) wheel also generates a traction force of 2 N. Then, what will be the forward and lateral direction forces? Assume $d=0.5m$
- b) What are proprioceptive and exteroceptive sensors for a mobile robot?
- c) What is Probabilistic, Map-Based Localization? Explain with the block diagram
- Unit – III**
7. a) Describe the bug algorithm and how it is used for obstacle avoidance in mobile robotics

- b) How does the Dijkstra's algorithm work for path planning, and how does it guarantee optimality in finding the shortest path between two points? Find the shortest path from 'S' to 'G' using the Dijkstra's algorithm for the graph representing a map for a mobile robot in Figure.Q7(b).

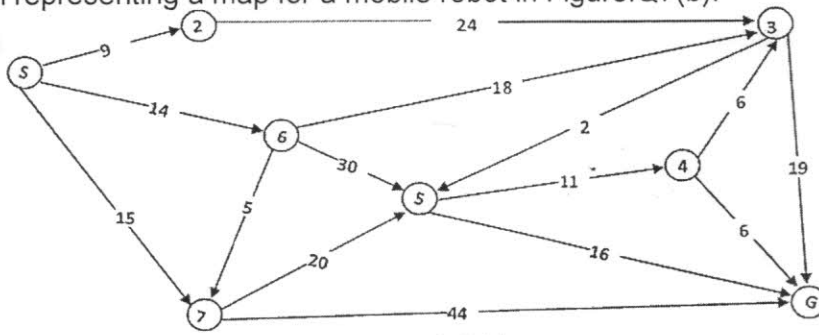


Figure. Q.7(b)

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8. a) How does the Voronoi diagram method work for path planning, and how is it used to generate paths that maximize distance from obstacles?
- b) Given the graph representing a map for a mobile robot in figure.Q8(b), where nodes correspond to waypoints, and edges represent possible paths with their associated travel costs, employ the A* search algorithm to find the most cost-efficient route for the robot to travel from the start node S to the goal node G.

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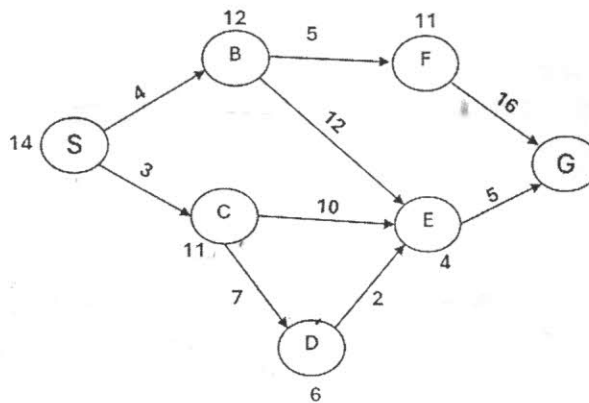


Figure. Q.8(b)

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
